

AI Usage Report

Course: SMD

Assignment: 01

Student Name: Saad Abdullah

Registration No.: 23i-3045

Date: 3/9/26

1. AI Tool(s) Used

List the AI tools used during the development of your application.

Example: ChatGPT, Gemini, GitHub Copilot

Response:

Google Gemini (via Antigravity IDE)

2. Purpose of AI Usage

Explain what you used AI for during the development of your application.

Examples:

- Understanding React/React Native concepts
- Debugging errors
- Generating or improving UI ideas
- Implementing charts using react-native-chart-kit
- Understanding unfamiliar code
- Improving code structure

Response:

I utilized AI primarily as a pair-programming assistant to build a robust, state-driven React Native application. Specifically, I used it for:

- Architecting a centralized state management system using React Hooks (useState) without relying on complex external navigation or database libraries.
- Designing a premium, modern UI incorporating "Glassmorphism" techniques using expo-blur.
- Implementing complex mathematical logic for dynamic grade weightage calculations and predictive scaling.
- Debugging React Native specific edge cases (e.g., UI rendering errors and TextInput state parsing bugs).

3. Important Prompts Used

Provide the important prompts that you gave to the AI tool.

Prompt 1:

I have an assignment to build a mobile application using React Native and Expo. Please review the assignment requirements document. Plan and implement this at the best level possible following all the requirements very strictly. Choose the 'Student Grades & Attendance Planner' problem to implement. Do not use external navigation libraries; manage the screens using state variables. Follow standard industry practices and ensure the UI looks premium.

Prompt 2:

I want to add proper deep functionality and innovative features. Make sure there is an option to record both 'Present' and 'Absent' using a Calendar date picker. For attendance, calculate exactly how many more classes need to be attended to reach an 80% threshold. Split the app into two separate modules: 'Attendance' and 'Grades Planner', keeping Grades Planner as the main focus. Inside Grades Planner, each course should have assessments like Quizzes (10%), Assignments (10%), Sessionals (15%), etc. Calculate the real-time grade status indicating Pass (≥ 50) or Fail.

Prompt 3:

Handle this edge case: allow adding multiple sub-items inside an assessment (like adding a 3rd Quiz). The assessment's overall weightage must remain the same, and the individual sub-items must automatically divide that weightage proportionally based on their total marks. Also, add a 'Predict Pass Requirement' toggle that only works if the total weightage is exactly 100%. If a student is failing, calculate the exact scaling factor or absolute marks needed to pass. If no realistic factor can save them, explicitly show that a factor won't work.

4. AI-Generated Output

Briefly describe the code, solution, or suggestion provided by the AI.

Response:

The AI generated the structural boilerplate for App.js, AttendanceDashboard.js, and GradePlannerDetails.js. It provided the core mathematical algorithms for the grading calculations (using JavaScript `.reduce()` to sum sub-item fractions) and suggested using `react-native-safe-area-context` to handle modern mobile screen notches.

5. Changes Made by Me

Explain what you changed, modified, or improved in the AI-generated solution.

Response:

I actively guided and restricted the AI's logic to match specific assignment requirements. For example, I modified the grading algorithm to ensure the "Predict Pass Requirement" feature is strictly locked unless the total assessment weightage equals exactly 100. I also dictated the specific UI/UX flow, enforcing standard industry practices (e.g., adding confirmation alerts, replacing harsh red icons with subtle UI elements, and ensuring no navigation libraries were used to artificially inflate the project scope).

6. My Understanding

Explain the important part(s) of the AI-generated code in your **own words**.

Response:

The application relies on a "Single Source of Truth" architecture. The `courses` array in App.js acts as the central state. Child components like GradePlannerDetails receive this data alongside a `setCourses` function. To calculate absolute marks, the code iterates over each assessment's sub-items. It divides the obtained marks by the total marks, and multiplies that fraction by the overall assessment weightage. Because the state is held in React's in-memory RAM, it resets on reload, effectively demonstrating pure frontend data manipulation.

7. Verification and Testing

Explain how you tested the AI-generated solution.

Mention:

- Whether the code worked correctly
- Errors encountered
- How you fixed the errors
- Any changes required after testing

Response:

I tested the application on an Android Emulator via Expo Go and the Expo Web interface. During testing, I encountered several issues:

- **SyntaxError (Red Screen):** The compiler failed due to an unescaped `<=` symbol inside JSX text. I fixed this by properly wrapping the symbol in JSX brackets `{ "<=" }`.
- **Deprecation Warnings:** I encountered warnings related to `SafeAreaView` and `DateTimePicker`. I fixed these by migrating to `react-native-safe-area-context` and updating the `onChange` props to `onValueChange`.
- **Decimal Input Bug:** I noticed decimal points were disappearing while typing. I resolved this by storing input values as raw strings in the React state and strictly executing `parseFloat()` only when mathematical calculations were occurring.

8. Reflection

What did you learn from using AI during this assignment?

Response:

I gained a deep understanding of how to manage deeply nested data structures (Arrays of Objects within Objects) using React State. I learned how complex it can be to handle raw mobile inputs (like numeric keyboards stripping floating-point decimals) and how to design premium interfaces using blurring techniques. Overall, I learned how to effectively direct an AI to assist with tedious boilerplate while retaining full architectural control over the application's logic.

Student Declaration

I confirm that I have used AI tools only as a development assistant and that I understand the code submitted as part of this assignment. I am able to explain and demonstrate the functionality of my application.

Student Name: Saad Abdullah

Signature: 

Date: 3/9/26